# Curriculum Vitae – Stefano Meza Anzules


**Personal Information**

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**Profile**

Physicist specializing in data science, with a Master's degree in Physics of Data and a strong foundation in statistical modelling, numerical methods, and scientific computing. Experienced in developing Python-based workflows to analyze complex scientific and applied datasets, using statistical analysis and machine learning across industrial sensor data, astronomical images, audio, and text.

**Education**

**Master's Degree in Physics of Data, University of Padova, Italy** [2022 – 2026]
Final grade: **104/110**
Thesis: *Detection and classification of objects in Metis solar corona images using deep learning.*
Relevant coursework: Machine Learning, Deep Learning and Neural Networks, Natural Language Processing, High-Dimensional Data, Statistical Mechanics, and Models of Theoretical Physics.

**Bachelor's Degree in Physics, Yachay Tech University, Ecuador** [2015 – 2019]
Final grade: **8.7/10**
Thesis: *Study of the scalar power spectrum in the generalized Starobinsky model.*
Relevant coursework: Computational Physics, Numerical Methods, Statistical Mechanics, Quantum Mechanics, and Mathematical Modelling.

**Experience**

- **Data Science Consultant**
  *GRAVITAD Systems S.L.* [Jan 2025 – Jan 2026]
  Contributed to technical and scientific consulting projects involving data processing and analysis. Used Python to support computational work and prepared technical documentation and reports.
- **Research Intern**
  *Istituto di Fotonica e Nanotecnologie (CNR-IFN), Padova, Italy* [Oct 2024 – Jul 2025]
  Developed image-processing and computer-vision methods to detect and characterize candidate objects—including possible sungrazing comets, asteroids, and other transient bodies—in more than 150,000 Metis images from ESA's Solar Orbiter mission. The work included feature extraction, clustering, temporal association, trajectory analysis, and a YOLO-based object-detection prototype.

**Selected Projects**

- **Restaurant Search and Recommendation with RAG**
  Built a retrieval-augmented system using data for 556 restaurants and 2,545 reviews. Combined semantic retrieval, Chroma vector search, metadata-based reranking, and a local language model.
- **Predictive Maintenance with PySpark**
  Analyzed approximately 142 million industrial sensor records. Engineered hourly features and compared logistic-regression models with persistence and constant-prediction baselines using chronological evaluation.
- **Generative Models: VAE and Diffusion**
  Implemented a convolutional VAE, unconditional DDPM, and class-conditional DDPM in PyTorch. Compared generated samples, sampling time, and classifier-based class agreement on MNIST and Fashion-MNIST.

## Publications

- **Meza, S.**, Altamirano, D., Mughal, M. Z. & Rojas, C. "Numerical analysis of the generalized Starobinsky inflationary model." *International Journal of Modern Physics D*, **30**, 2150062 (2021). DOI: 10.1142/S0218271821500620.
- **Meza, S.**, Chioetto, P., Moualla, L., Casini, C., Corso, A., Frassetto, F., Naletto, G., Zuppella, P. & Da Deppo, V. "Detection and classification of objects in Metis solar corona images using deep learning." In: *Applications of Machine Learning 2025*, Proc. SPIE **13606**, 315–320 (2025). DOI: 10.1117/12.3063859.

## Technical Skills

**Programming and data:** Python, R, SQL, NumPy, pandas, SciPy, PySpark.

**Machine learning:** scikit-learn, PyTorch, TensorFlow/Keras, classification, clustering, CNNs, RNNs, object detection, generative models, RAG, embeddings, vector databases, large language models.

**Scientific computing:** Numerical simulation, Bayesian inference, scientific image processing, astronomical data analysis, SPICE.

**Tools:** Linux, Git, GitHub, Jupyter, Docker, APIs, LaTeX.

**Languages:** Spanish (native), English (C1), Italian (B2).